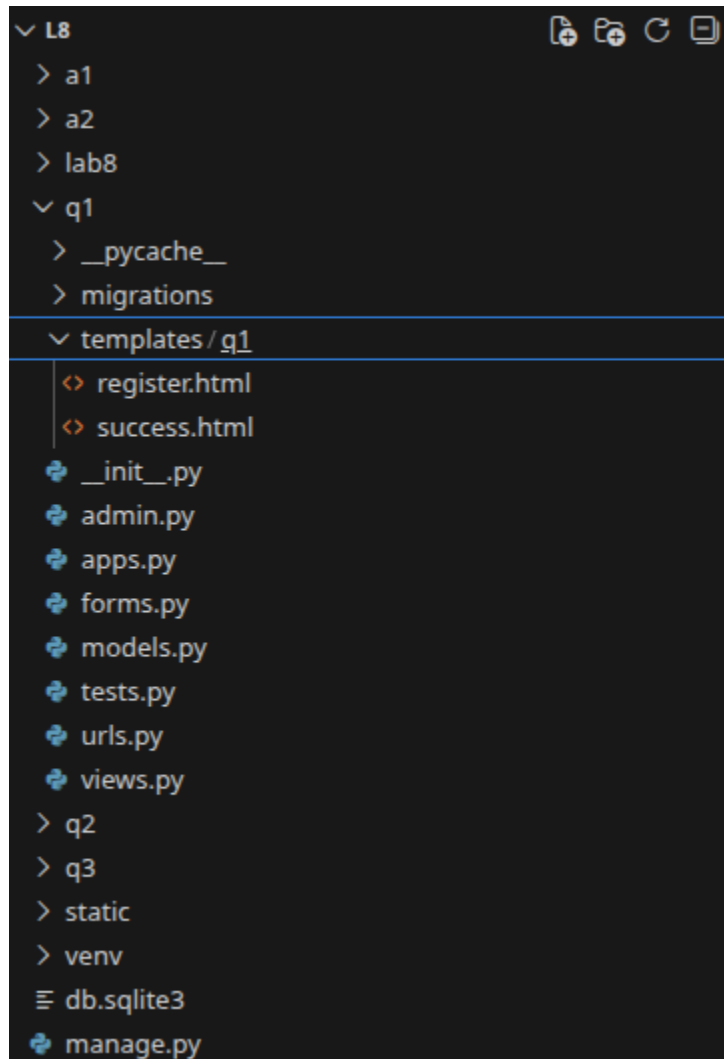


Form Processing Using Django – Part 2

Folder Structure:



Settings.py:

```
INSTALLED_APPS = [  
    'django.contrib.admin',  
    'django.contrib.auth',  
    'django.contrib.contenttypes',  
    'django.contrib.sessions',  
    'django.contrib.messages',  
    'django.contrib.staticfiles',  
    'q1',  
    'q2',  
    'q3',  
    'a1',  
]
```

```
'a2',  
]
```

Urls.py:

```
urlpatterns = [  
    path('admin/', admin.site.urls),  
    path('q1/', include('q1.urls')),  
    path('q2/', include('q2.urls')),  
    path('q3/', include('q3.urls')),  
    path('a1/', include('a1.urls')),  
    path('a2/', include('a2.urls')),  
]
```

Q1.

Register.html:

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
<meta charset="UTF-8">  
<meta name="viewport" content="width=device-width, initial-scale=1.0">  
<title>Register - Q1</title>  
<link rel="stylesheet" href="/static/css/q1.css">  
</head>  
<body>  
<div class="container q1-container">  
<h1>Register</h1>  
<form method="POST">  
    {% csrf_token %}  
    <div class="form-group">  
        <label for="{{ form.username.id_for_label }}">{{ form.username.label }} <span  
class="required">*</span></label>  
        {{ form.username }}  
        {% if form.username.errors %}  
        <span class="error">{{ form.username.errors }}</span>  
        {% endif %}  
    </div>  
  
    <div class="form-group">  
        <label for="{{ form.password.id_for_label }}">{{ form.password.label }}</label>  
        {{ form.password }}  
    </div>  
  
    <div class="form-group">  
        <label for="{{ form.email.id_for_label }}">{{ form.email.label }}</label>  
        {{ form.email }}
```

```

</div>

<div class="form-group">
<label for="{{ form.contact.id_for_label }}">{{ form.contact.label }}</label>
{{ form.contact }}
</div>

<button type="submit" class="btn-submit">Submit</button>
</form>
</div>
</body>
</html>

```

Success.html:

```

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Success - Q1</title>
<link rel="stylesheet" href="/static/css/q1.css">
</head>
<body>
<div class="container q1-container">
<h1>Welcome {{ username }}</h1>
<div class="success-info">
{% if email %}
<p><strong>Email:</strong> {{ email }}</p>
{% endif %}
{% if contact %}
<p><strong>Contact Number:</strong> {{ contact }}</p>
{% endif %}
</div>
<a href="{% url 'q1:register' %}" class="btn-link">Register Again</a>
</div>
</body>
</html>

```

Forms.py:

```

from django import forms

class RegisterForm(forms.Form):
username = forms.CharField(max_length=100, label='Username',
widget=forms.TextInput(attrs={'class': 'form-input', 'placeholder': 'Enter username'}))
password = forms.CharField(max_length=100, label='Password', required=False,

```

```

widget=forms.PasswordInput(attrs={'class': 'form-input', 'placeholder': 'Enter password'}))
email = forms.EmailField(label='Email', required=False,
widget=forms.EmailInput(attrs={'class': 'form-input', 'placeholder': 'Enter email'}))
contact = forms.CharField(max_length=20, label='Contact Number', required=False,
widget=forms.TextInput(attrs={'class': 'form-input', 'placeholder': 'Enter contact number'}))

```

Urls.py:

```

from django.urls import path
from . import views

app_name = 'q1'

urlpatterns = [
    path('register/', views.register, name='register'),
    path('success/', views.success, name='success'),
]

```

Views.py

```

from django.shortcuts import render, redirect
from .forms import RegisterForm

def register(request):
    if request.method == 'POST':
        form = RegisterForm(request.POST)
        if form.is_valid():
            request.session['username'] = form.cleaned_data['username']
            request.session['email'] = form.cleaned_data['email']
            request.session['contact'] = form.cleaned_data['contact']
            return redirect('q1:success')
        else:
            form = RegisterForm()
            return render(request, 'q1/register.html', {'form': form})

def success(request):
    username = request.session.get('username', 'User')
    email = request.session.get('email', "")
    contact = request.session.get('contact', "")
    return render(request, 'q1/success.html', {
        'username': username,
        'email': email,
        'contact': contact
    })

```

Screenshots:

The screenshot shows a web browser window with the title 'Register - Q1'. The address bar shows 'localhost:8000/q1/register/'. The page contains a registration form with the following fields and values:

- Username ***: wpl
- Password**: (masked)
- Email**: wpl123@gmail.com
- Contact Number**: 1234567890

A blue 'Submit' button is located at the bottom of the form.

The screenshot shows a web browser window with the title 'Success - Q1'. The address bar shows 'localhost:8000/q1/success/'. The page displays a success message:

Welcome wpl

Email: wpl123@gmail.com
Contact Number: 1234567890

A blue 'Register Again' button is located at the bottom of the message box.

Q2.

Results.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Results - Q2</title>
<link rel="stylesheet" href="/static/css/q2.css">
</head>
<body>
```

```

<div class="container q2-container">
<h1>Voting Results</h1>
<div class="results">
<div class="result-item">
<label>Good:</label>
<span class="percentage">{{ good_percent }}%</span>
<span class="votes">({{ good_votes }} votes)</span>
</div>
<div class="result-item">
<label>Satisfactory:</label>
<span class="percentage">{{ satisfactory_percent }}%</span>
<span class="votes">({{ satisfactory_votes }} votes)</span>
</div>
<div class="result-item">
<label>Bad:</label>
<span class="percentage">{{ bad_percent }}%</span>
<span class="votes">({{ bad_votes }} votes)</span>
</div>
<div class="result-item total">
<label>Total Votes:</label>
<span>{{ total }}</span>
</div>
</div>
<a href="{% url 'q2:vote' %}" class="btn-link">Vote Again</a>
</div>
</body>
</html>

```

Vote.html:

```

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Vote - Q2</title>
<link rel="stylesheet" href="/static/css/q2.css">
</head>
<body>
<div class="container q2-container">
<h1>How is the book ASP.NET with C# by Vipul Prakashan?</h1>
<form method="POST">
{% csrf_token %}
<div class="form-group">
<label>
<input type="radio" name="choice" value="good" required>

```

```

Good
</label>
</div>
<div class="form-group">
<label>
<input type="radio" name="choice" value="satisfactory" required>
Satisfactory
</label>
</div>
<div class="form-group">
<label>
<input type="radio" name="choice" value="bad" required>
Bad
</label>
</div>
<button type="submit" class="btn-submit">VOTE</button>
</form>
</div>
</body>
</html>

```

Urls.py:

```

from django.urls import path
from . import views

app_name = 'q2'

urlpatterns = [
    path('vote/', views.vote, name='vote'),
    path('results/', views.results, name='results'),
]

```

Views.py

```

from django.shortcuts import render, redirect
from django.http import JsonResponse

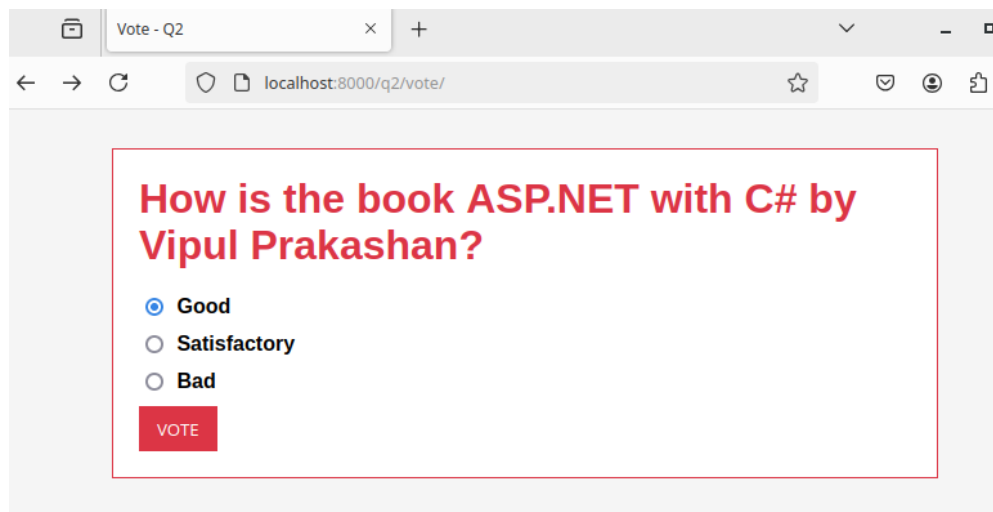
# Store votes in session
def vote(request):
    if request.method == 'POST':
        choice = request.POST.get('choice')
        if choice in ['good', 'satisfactory', 'bad']:
            votes = request.session.get('votes', {'good': 0, 'satisfactory': 0, 'bad': 0})
            votes[choice] = votes.get(choice, 0) + 1
            request.session['votes'] = votes
        return redirect('q2:results')

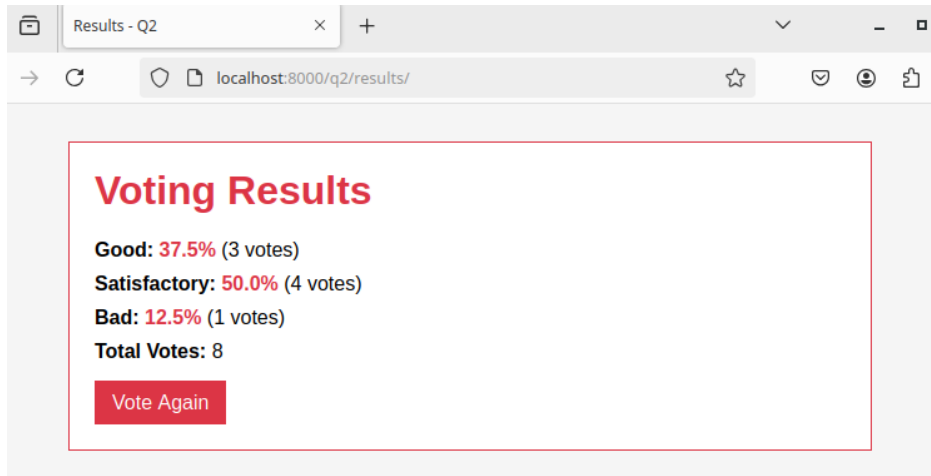
```

```
return render(request, 'q2/vote.html')

def results(request):
    votes = request.session.get('votes', {'good': 0, 'satisfactory': 0, 'bad': 0})
    total = sum(votes.values())
    if total > 0:
        good_percent = (votes['good'] / total) * 100
        satisfactory_percent = (votes['satisfactory'] / total) * 100
        bad_percent = (votes['bad'] / total) * 100
    else:
        good_percent = satisfactory_percent = bad_percent = 0
    return render(request, 'q2/results.html', {
        'good_votes': votes['good'],
        'satisfactory_votes': votes['satisfactory'],
        'bad_votes': votes['bad'],
        'good_percent': round(good_percent, 2),
        'satisfactory_percent': round(satisfactory_percent, 2),
        'bad_percent': round(bad_percent, 2),
        'total': total
    })
```

Screenshots:





Q3.

Calculate.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>CGPA Calculator - Q3</title>
<link rel="stylesheet" href="/static/css/q3.css">
</head>
<body>
<div class="container q3-container">
<h1>CGPA Calculator</h1>
<form method="POST">
{% csrf_token %}
<div class="form-group">
<label for="{{ form.name.id_for_label }}">{{ form.name.label }}</label>
{{ form.name }}
</div>

<div class="form-group">
<label for="{{ form.marks.id_for_label }}">{{ form.marks.label }}</label>
{{ form.marks }}
</div>

<button type="submit" class="btn-submit">Calculate</button>
</form>
</div>
</body>
</html>
```

Result.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Result - Q3</title>
<link rel="stylesheet" href="/static/css/q3.css">
</head>
<body>
<div class="container q3-container">
<h1>CGPA Result</h1>
<div class="result-info">
<p><strong>Student Name:</strong> {{ name }}</p>
<p><strong>Total Marks:</strong> {{ marks }}</p>
<p><strong>CGPA (Marks/50):</strong> {{ cgpa }}</p>
</div>
<a href="{% url 'q3:calculate' %}" class="btn-link">Calculate Again</a>
</div>
</body>
</html>
```

Forms.py:

```
from django import forms

class CGPAForm(forms.Form):
    name = forms.CharField(max_length=100, label='Student Name',
        widget=forms.TextInput(attrs={'class': 'form-input', 'placeholder': 'Enter name'}))
    marks = forms.DecimalField(max_digits=5, decimal_places=2, label='Total Marks',
        widget=forms.NumberInput(attrs={'class': 'form-input', 'placeholder': 'Enter marks', 'step': '0.01'}))
```

Urls.py:

```
from django.urls import path
from . import views

app_name = 'q3'

urlpatterns = [
    path('calculate/', views.calculate, name='calculate'),
    path('result/', views.result, name='result'),
]
```

Views.py

```
from django.shortcuts import render, redirect
from .forms import CGPAForm
```

```
def calculate(request):
    if request.method == 'POST':
        form = CGPAForm(request.POST)
        if form.is_valid():
            name = form.cleaned_data['name']
            marks = form.cleaned_data['marks']
            cgpa = float(marks) / 50
            request.session['student_name'] = name
            request.session['total_marks'] = float(marks)
            request.session['cgpa'] = round(cgpa, 2)
            return redirect('q3:result')
        else:
            form = CGPAForm()
            return render(request, 'q3/calculate.html', {'form': form})

def result(request):
    name = request.session.get('student_name', 'Student')
    marks = request.session.get('total_marks', 0)
    cgpa = request.session.get('cgpa', 0)
    return render(request, 'q3/result.html', {
        'name': name,
        'marks': marks,
        'cgpa': cgpa
    })
```

Screenshots:

